Mert Kilickaya

Senior AI Research Engineer

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Summary

Senior AI Research Engineer focused on foundation models and autonomous vision. Combines self-supervised learning, continual learning, and vision-language learning to develop agentic vision systems that improve through experience. Research aims to let models select useful data, derive supervision, and adapt to changes in the real world.

Experience

Senior Research Engineer September 2026 – Present
Greyparrot, London, UK (remote from the Netherlands)

- Develop and evaluate visual foundation models for autonomous perception, with a focus on transferable representations.
- Research agentic vision systems that select useful observations and derive training supervision with reduced manual intervention.
- Investigate continual adaptation from new visual observations, with a focus on distribution shifts and catastrophic forgetting.

Deep Learning Researcher 2023 – October 2026
Agendia, Amsterdam, Netherlands

- Led the design and development of a deep-learning-based biomarker for breast cancer recurrence, owning model architecture, validation strategy, and regulatory-facing documentation under IVDR constraints.
- Built and deployed end-to-end multiple-instance learning pipelines over tens of thousands of whole-slide images, covering data curation, training, evaluation, and containerized inference in PyTorch.
- Drove technical decision-making across the full ML lifecycle, collaborating with clinicians and biologists to align model behavior with clinical objectives and deployment requirements.
- Contributed to a filed patent, IVDR submission work, and research presented at ESMO 2026.

Postdoctoral Researcher 2022 – 2023
Eindhoven University of Technology, Netherlands

- Conducted research on continual pre-training of foundation models under distribution shift, focusing on mitigating catastrophic forgetting in long-running self-supervised training regimes.
- Defined research directions and acted as a de facto co-supervisor for 5+ PhD and MSc students, leading to publications at ECCV and BMVC.

Research Intern 2020 – 2021
Huawei Visual Search Lab, Helsinki, Finland

- Improved large-scale mobile visual search systems under strict latency and memory constraints for on-device and near-device deployment.
- Developed text-interactive visual search models optimized for efficiency, resulting in an approved US patent.

PhD Researcher 2017 – 2022
Qualcomm Labs, University of Amsterdam, Netherlands

- Developed deep vision models for visual search, detection, and recognition.
- Contributed to research that resulted in 4 granted US patents and multiple top-tier publications.

Education

PhD in Deep Learning 2017 – 2022
University of Amsterdam, Netherlands
Thesis: *Contextual Understanding of Visual Interactions*
Supervisor: *Prof. Arnold Smeulders*

MSc in Computer Vision 2013 – 2016
Hacettepe University, Turkey
Thesis: *Visual Importance with Applications to Vision and Language*
Supervisor: *Prof. Nazli Ikizler-Cinbis*

Selected Publications

(h-index: 7, Citations: 370)
Top-tier venues including ECCV, BMVC, WACV, and EACL

- AI Driven Digital Pathology for the 70 Gene Recurrence Signature: Development and Validation in Early Stage Breast Cancer (ESMO 2026, Oral)
- HyTAS: A Transformer Architecture Search Benchmark (ECCV 2024)
- Locality-Aware ViTs for Hyperspectral Imaging (BMVC 2023)
- Are Labels Needed for Incremental Instance Learning? (CVPRW 2023, Oral)
- Structured Visual Search via Composition-Aware Learning (WACV 2021)
- HOI Detection via Weak Supervision (BMVC 2021)
- Re-evaluating Automatic Metrics for Image Captioning (EACL 2017, Oral)

Patents

- Visual Search via Conversational Interaction (Huawei, 2022)
- Network for Interacted Object Localization (Qualcomm, 2022)
- Context-driven Learning of Human-Object Interactions (Qualcomm, 2021)
- Subject-Object Interaction Recognition Model (Qualcomm, 2022)

Technical Skills

Area	Expertise
Research	Visual foundation models, self-supervised learning, continual learning, vision-language learning
Autonomous vision	Data selection, supervision discovery, adaptation to distribution shifts, temporal learning
Computer vision	Object detection, classification, tracking, visual retrieval, whole-slide image analysis
Model development	PyTorch, transformer fine-tuning, multiple-instance learning, model evaluation
Engineering	Python, Docker, GPU training and inference, experiment tracking

Service & Awards

Outstanding Reviewer: ECCV 2024

Reviewer: CVPR 2025, ECCV 2024, ICML, NeurIPS, ICLR 2023–2024, EMNLP 2021–2023